

# DBMS

## EXERCISE HUB

### EXERCISE SOLUTIONS

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Unit = 1

## Database and Database Users :

1) What is database? How it is differ from traditional file system?

→ A database is organised collection of data that is stored electronically and managed using a software called a Database management System (DBMS).

| Basis            | Database System                                                | Traditional File System                 |
|------------------|----------------------------------------------------------------|-----------------------------------------|
| Meaning          | Data is stored in a structured and centralized way using DBMS. | Data is stored in separate files        |
| Data Redundancy  | Very low redundancy                                            | High redundancy (duplication data)      |
| Data consistency | Better consistency                                             | Limited consistency problems may occurs |
| Security         | Strong security & user permissions                             | Limited security                        |
| Examples         | Banking System, School management System                       | Separate text files, Excel files.       |



2) What are the advantages of a database system?  
What are the various cost and risk factors involved in implementing a database system?  
⇒ Advantages of database system

1 Reduced Data Redundancy

→ Duplicate data is minimized because data is stored centrally.

2 Improved Data consistency

→ Since duplication is reduced, data remains accurate and consistent

3 Better Data Security

→ Database systems provide password, access control and permissions to protect data.

4 Data Sharing

→ Multiple user can access the same database at the same time.

5 Backup and Recovery

→ DBMS provides facilities for backup and restoring lost data.



# # cost and Risk factors In implementing a Database System.

→ Although database systems have many advantages there are also some costs and risks involved.

## A cost factors:

### 1) Hardware cost

→ Powerful computers, storage devices and networking equipment may be required.

### 2) Software cost

→ DBMS software license can be expensive.  
e.g

Oracle Database

Microsoft SQL Server

### 3) Training cost

→ Employees and database administrators need proper training.

### 4) Maintenance cost

→ Regular updates, monitoring and maintenance are needed.



## B Risk Factors :

### 1) Data Security Risk

→ Unauthorized users or hackers may access sensitive informations if security is weak.

### 2. System Failure Risk

→ Hardware or Software failure can affect the entire database system.

### 3 Data Loss Risk

→ Improper backup may lead to permanent data loss.

### 4. Complexity

→ Database systems are more complex than traditional file systems and require skilled professionals.

### 5 High Dependency on DBMS

→ If the DBMS fails, many applications may stop working.



3) What do you understand by the term data abstraction?

→ Data abstraction is the process of hiding unnecessary details from users and showing only the essential information.

4) Write down advantages and disadvantages of Flat File System.

→

| Advantage of Flat File System       | Disadvantage of Flat File System        |
|-------------------------------------|-----------------------------------------|
| i) Simple and easy to use           | i) Data redundancy occurs               |
| ii) Low cost                        | ii) Data inconsistency may occur        |
| iii) Easy to set up                 | iii) Poor security                      |
| iv) Suitable for small applications | iv) Difficult data sharing              |
| v) Portable and easy to transfer    | v) No proper relationship between data. |

5) What are the application areas of Flat File System? Explain.

→ A Flat File System is used where data is small, simple and does not need complex relationships or heavy processing. It is common in basic storage tasks and small applications.



# i) Personal Data Storage.

→ used by individual to store simple information  
e.g

Contact list in a phone  
personal notes in a text file.

# ii) Small Business Records.

→ small shops or businesses use flat files to maintain basic records.

e.g Customer list

# iii) Educational purposes

→ used in school and colleges for small projects or assignments.

e.g attendance records for a small class.

# iv) Temporary Data Storage

→ used for short-term or temporary storage of data.

e.g Cache files, Temporary processing files.

# v) log files in systems

→ Operating systems and applications store logs in flat files.

e.g error log  
Server log



6) Draw the various area in which database system is used.

→ Database Systems are used in almost every field where data needs to be stored, managed and accessed efficiently. They help organizations handle large amounts of information in an organized way.

i) Banking System

→ Banks use database to manage customer accounts and transactions.

ii) Education System

→ Schools, colleges and universities use database to manage academic data.

iii) Healthcare System

→ Hospitals use databases to manage patient information.

iv) Railway and Airline Reservation System

→ Used for booking and managing tickets.

v) Government Organizations

→ Government departments use database for public records  
e.g. citizen ID cards.



7) Who is a DBA? List the various responsibilities of DBA.

→ A DBA (Database Administrator) is a person who is responsible for managing, maintaining and controlling a database system.

### Responsibilities of DBA:

i) Database Installation and Setup

→ The DBA installs and configures the database software in the system.

e.g.

Setting up systems like MySQL

ii) User management

→ The DBA creates users and controls their access rights.

e.g.

Giving access to teacher to enter marks

iii) Security management

→ Ensure that only authorized users can access the database.

Include,

- Password Protection
- Access Control
- Encryption



- iv) Backup and Recovery  
 → The DBA regularly takes backup of data and restores it in case of failure.  
 e.g. Recovering data after system crash.
- v) Software and Hardware maintenance  
 → Ensures that database hardware and software are working properly.
- vi) Troubleshooting  
 → Solves database errors and technical problems.
- 8) What are the advantages of different types of database users who interact with the database system?  
 → Different types of users interact with a database system based on their roles and level of access.
- 1) Database Administrator (DBA)  
 → A DBA manages and controls the entire database system.  
 Responsibilities:  
 - Security management  
 - Backup and recovery  
 - User access control  
 - Performance monitoring  
 e.g. Oracle Database, MySQL,



## 2) Database Designers

→ They design the structure of the database  
Responsibilities:

- create tables and relationships
- Define schema (structure)
- Plan how data is stored.

## 3) Application programmers

→ These user write programs to interact with the database

Responsibilities

- Develop application using database queries
- connect software with DBMS.

eg

Banking app developers

E-commerce website developers

## 4) End users

→ These are the people who use the database indirectly or directly to get information

They are further divided into

### a) Naive users (simple users)

→ Naive user are user who have little or no knowledge about database system. They are predefined application to access the database

eg ATM, machine, Railway ticket Booking



## b) Sophisticated users

→ Sophisticated users have good knowledge of database and can write complex queries using SQL or other database language.

eg Data analysts  
Engineers.

Software.

my SQL,

## c) Specialized users.

→ Specialized users are experts who develop special database application for scientific research, artificial intelligence, multimedia system or engineering purposes.

eg AI Developer

Scientists

CAD system designers.

## 9) What are the responsibilities of DBA? Explain.

→ Same as QN 7.

## 10) Describe different types of database users.)

→ Details explained in QN 8.